HBase:The Data Base for Hadoop

Modern living generates large volumes of data which have to be stored, interpreted and analysed for various purposes. The Java based Hadoop File System (HDFS) provides scalable and reliable data storage. However, HDFS is not a database nor does it have random read/write access to Big Data. HBase – a distributed, versioned, non-relational database – comes to the rescue. Learn how to install HBase in this article.

In today's world, data is knowledge and is increasing each day by leaps and bounds. The data versus time graph shows exponential increase. We can't erase this data, as it is mined to fetch very useful information. For example, Sensex data is stored and read to make predictive analyses about stocks. The 'claims' data of the insurance companies is analysed to detect fraudulent claims. Even the DNA data is being studied to find cures for genetic diseases. Hadoop is the open source technology that handles this huge amount of data. In Hadoop, data is stored in HDFS in the form of data chunks, which are distributed across the nodes of the cluster. But the basic problem of Hadoop is that it is not a database, and does not have the facility to write many times, and that, too, in real-time. Hadoop is used for batch processing, and works on the principle of 'write once, read many times'. HBase has been developed to address this problem.

HBase is a distributed column-oriented database built on top of HDFS. It is the Hadoop application to use when you require real-time read/write random access to very large data sets. The HBase project was started by Chad Walters and Jim Kellerman towards the end of 2006 at the company called Powerset. It was modelled after Google's Big Table. HBase is not relational and does not support SQL, but unlike relational databases it can host very large, sparsely populated tables on clusters made of commodity hardware.

In HBase, applications store data in labelled tables, where tables are made of rows and columns. The intersections of row and column coordinates are versioned, and by default their version is time stamped and auto-assigned by HBase at the time of cell creation. The table row keys are byte arrays, so anything can serve as the row key. Table rows are sorted by row keys, where the sort is byte ordered. Row columns are grouped into column families, which have a common prefix with column family information as name, email and password. A table column family must be specified up front as a part of the table schema definition, but new column family members can be added on demand. Tables are automatically portioned horizontally by HBase into regions. Each region comprises a subset of table rows; the subset consists of elements where the first row is taken into account and the last row is excluded. The last row acts as the first row of the other region. Initially, all the data is stored in one region server, but as the data grows and crosses a configurable size threshold, it splits to two region servers. In the context of locking, the